

A House Divided: Modern Heterogeneous-Effects Estimators and the Contested Mortality Effects of Hospital Desegregation, 1959–1988

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Abstract

We revisit the contested Almond–Chay–Greenstone (2006) versus Anderson–Charles–Rees (2024) adjudication on Title VI hospital desegregation and Black postneonatal mortality using five staggered-DiD estimators on an eleven-state Confederate-South panel extended back to 1959 from NCHS microdata and audited state-vital-statistics natality. On the extended panel, all five estimators reject zero: TWFE -1.85 , Callaway–Sant’Anna -5.09 , Borusyak–Jaravel–Spiess -2.41 , de Chaisemartin–D’Haultfœuille -4.07 , Sun–Abraham -2.10 per 1,000 live births. The original disagreement decomposes into a cohort-anchoring component (the dominant 1967 cohort has no pre-period in post-1968 public data) and a forbidden-comparison component in TWFE.

Keywords: Title VI, hospital desegregation, Black infant mortality, staggered difference-in-differences, Callaway–Sant’Anna, Goodman-Bacon decomposition.

JEL codes: I14, I18, J15, N32.

1. Introduction

The 1964 Civil Rights Act was operationally enforced in southern hospitals through Medicare certification. Beginning July 1, 1966, the Department of Health, Education, and Welfare’s Office of Equal Health Opportunity (HEW–OEHO) required, as a condition of receiving federal Medicare reimbursement, that hospitals certify nondiscrim-

inatory treatment of Black patients and staff under Title VI [reynolds1997federal; smith1999healthcaredivided]. The campaign was, in Reynolds’s (1997) characterization, “one of the most successful civil-rights enforcement operations of the federal government.” Within eighteen months, 96 percent of the Deep South’s hospitals were certified; the remainder either closed or accepted formal segregation and forfeited federal funds.

Twenty years later, Almond, Chay, and Greenstone (2006) [hereafter ACG] used the NBER Compressed Mortality File and the staggered 1966–1968 county-level certification rollout to argue that this campaign produced a large, persistent decline in Black postneonatal mortality across the rural South. The ACG paper, circulated as MIT Working Paper 07-04 in 2007 but never published in a peer-reviewed journal, became the canonical empirical reference for the proposition that civil-rights enforcement saved tens of thousands of Black infant lives in the late 1960s. It is cited in virtually every survey of post-1960s mortality declines and has been influential in establishing the broader narrative that Black–White health gaps in the United States closed substantively only when federal civil-rights enforcement gave Black Americans physical access to white-only health-care infrastructure.

Anderson, Charles, and Rees [hereafter ACR] revisited the question in a 2024 paper now forthcoming at the *Review of Economics and Statistics*. ACR (a) compiled the underlying hospital-certification dates directly from the annual *Journal of the American Hospital Association* “Guide Issue” rather than relying on the secondary sources that ACG had used; (b) coded county-level “access” using four different bed-coverage thresholds (any certification, ≥ 25 percent, ≥ 50 percent, ≥ 75 percent of pre-1965 hospital beds); and (c) restricted attention to the five Deep-South states most exposed to the certification campaign. Their headline conclusion, on the ≥ 75 -percent bed-coverage binding (`cert_75`), is that the campaign had no statistically detectable effect on Black postneonatal mortality. ACR (2024) is the first careful re-examination of ACG’s headline claim and reports a *null* on essentially the same outcome.

The disagreement matters for two reasons. First, the foundational claim that civil-rights enforcement produced a large, measurable mortality dividend is uncomfortable to lose; the absence of a peer-reviewed Almond–Chay–Greenstone paper has meant that the field has cited the working-paper version for two decades as if it were settled. Second, the methodological texture of the disagreement is informative for a broader question: how should the discipline now read the large 2000s and 2010s difference-in-differences literature that relied on the staggered-TWFE estimator? Goodman-Bacon (2021), Callaway and Sant’Anna (2021), Sun and Abraham (2021), de Chaisemartin and D’Haultfœuille (2020), Borusyak, Jaravel, and Spiess (2024), and Rambachan and Roth (2023) have collectively documented that staggered-TWFE estimators can deliver biased estimates of average treatment effects when treatment timing varies, treatment effects are heterogeneous, or pre-trends are present. None of those papers has been applied side-by-side to the ACG–ACR question.

This paper does so. We construct a unified eleven-state Confederate-South county-year panel for the period 1968–1988 — the public-NBER-CMF window — and apply five staggered-DiD estimators (TWFE, Callaway–Sant’Anna, Borusyak–Jaravel–Spiess,

de Chaisemartin–D’Haultfœuille, and Sun–Abraham) to the same outcome (Black postneonatal mortality per 1,000 live births), the same treatment definition (the ACR `cert_75` binding), and the same sample frame. We then extend the panel backward to 1959 by transcribing pre-1968 state vital-statistics tabulations — the same data source that ACG (2006) used for their pre-period and that ACR (2024) acknowledged as essential for the dynamic event-study but did not themselves transcribe. The extended-panel estimates reveal whether the original disagreement was driven by the post-1968-only NBER CMF window or by something else.

Our contribution is methodological adjudication, not a new empirical claim about whether hospital desegregation reduced Black infant mortality. We document the *shape* of the estimator frontier on this canonical question — under what data window, under what estimator, the data favors the ACG reading; under what conditions the data favors ACR; and what the partial-identification frontier looks like under Rambachan–Roth bounds. We hope the exercise is useful both as a substantive adjudication of one of the most important questions in modern U.S. health-equity history and as a methodological case study for how to read older staggered-DiD literatures in light of the post-2018 econometric developments.

The paper proceeds as follows. Section 2 reviews the historical and econometric literature. Section 3 documents the data. Section 4 describes the five estimators and the robustness suite. Section 5 reports results. Section 6 discusses the partial-identification frontier and what would be needed to pin the magnitude. Section 7 concludes.

2. Background

2.1 The Title VI hospital-desegregation campaign — institutional and political history

The 1964 Civil Rights Act outlawed discrimination by recipients of federal funds, but its operative enforcement in southern hospitals required a separate instrument: the federal money attached to Medicare. Before 1966, southern hospitals received federal Hill-Burton construction grants without operative Title VI compliance. The 1946 Hill-Burton Hospital Survey and Construction Act explicitly authorized “separate but equal” facilities — Section 622(f) of the Act permitted federal funds to flow to hospitals that maintained racial separation provided the separate Black facilities were “of like quality.” Between 1946 and 1965, the federal Hill-Burton program funded approximately 100 segregated Black hospitals across the South and provided construction grants to hundreds of additional white-only facilities [[@smith1999healthcaredivided](#); [@reynolds1997federal](#)].

The constitutional challenge to Hill-Burton’s separate-but-equal clause culminated in the 1963 Fourth Circuit decision in *Simkins v. Moses H. Cone Memorial Hospital* (323 F.2d 959, 4th Cir. 1963), in which the court ruled that the Hill-Burton separate-but-equal provision was unconstitutional under the Equal Protection Clause and that hospitals receiving federal funds could not maintain racial segregation. The decision applied

to the Fourth Circuit states (NC, SC, VA, WV, MD) but not to the Fifth Circuit states (AL, FL, GA, LA, MS, TX). The Supreme Court denied certiorari in 1964 (376 U.S. 938), leaving Simkins as Fourth-Circuit-only precedent until the broader 1964 Civil Rights Act and the 1966 Medicare Act enforcement.

The operative enforcement instrument was Title VI of the 1964 Civil Rights Act applied through Medicare certification, beginning July 1, 1966. Under the Medicare Act of 1965 (Public Law 89-97), hospitals receiving Medicare reimbursement were required to certify nondiscrimination in admissions, room assignments, staff privileges, and physician practice. The certification process was operationally managed by the Department of Health, Education, and Welfare’s Office of Equal Health Opportunity (HEW–OEHO), established within the Public Health Service. HEW–OEHO compliance officers conducted on-site inspections of southern hospitals during February 1966 through October 1968, verifying that hospitals had eliminated racially-designated wards, integrated staff dining facilities, opened medical-staff appointments to Black physicians, and complied with the Title VI nondiscrimination requirements.

The political economy of the certification process was complex. Hospital trustees in the Deep South initially resisted certification — 214 hospitals in the five-state Deep South refused federal Medicare reimbursement rather than integrate. The economic pressure of forgoing Medicare reimbursement, however, proved decisive: within twelve months of the July 1, 1966 effective date, approximately 96 percent of Deep-South hospitals had certified. The laggards were predominantly small, rural facilities with limited Medicare-eligible patient bases and limited financial reserves; they either closed (~30 closures in 1966–1969) or reluctantly certified. The political-historical literature [[@reynolds1997federal](#); [@smith1999healthcaredivided](#); [@smith2005race](#); [@largent2018segregation](#)] documents that the certification process was conducted under substantial federal-state political tension, with state governors and southern members of Congress repeatedly attempting to block or delay HEW–OEHO compliance enforcement.

The HEW–OEHO compliance files themselves are archived at the National Archives and Records Administration as NARA Record Group 235, Series 5 (Health Equity Compliance Files, 1966–1972). Reynolds (1997) and Largent (2018) have separately conducted archival work in these files; their accounts of the operational compliance process — site inspections, formal compliance reviews, hospital-by-hospital certification negotiations — are the primary source basis for our understanding of cosmetic-compliance heterogeneity. Smith (1999, 2005) documents that many certified hospitals dual-listed wards as “integrated” while continuing de facto segregation through admission patterns, room-assignment defaults, and physician-practice restrictions. The `cert_75` binary indicator in our analytic panel averages over this substantial residual segregation.

The campaign’s operational success was measured at the federal level through the “Guide Issues” of the *Journal of the American Hospital Association*, published annually in February and October. The Guide Issue listed, for each hospital, whether it had been “certified for participation in the Health Insurance for the Aged (Medicare) Program by the Department of Health, Education, and Welfare” as of the publication’s reference date. Anderson, Charles, and Rees (2024) hand-coded the Guide Issues

for the five Deep-South states (AL, GA, LA, MS, SC) for 1967–1974 to reconstruct hospital-level certification dates, then aggregated to the county-of-nearest-hospital level under four alternative bed-coverage thresholds: any certification (`cert_any`), ≥ 25 percent of pre-1965 county hospital beds (`cert_25`), ≥ 50 percent (`cert_50`), and ≥ 75 percent (`cert_75`). The resulting county-year certification panel — reported in their Appendix Table B1 — is the analytic foundation for the modern reassessment of the ACG findings.

The dispersion in `cert_75` is meaningfully wider than in `cert_any`. In our merged panel, 350 of 415 Deep-South counties with assigned dates were certified in 1967 under `cert_75`; 43 in 1968; 17 in 1969–1971; and 5 in 1972 or later. Under `cert_any` (any hospital certified), the distribution is much more concentrated in 1966–1967. The dispersion in `cert_75` is exactly the variation that ACR (2024) argue powers the modern county-of-nearest-hospital identification strategy and that ACG (2006) treated as too modest to exploit.

2.2 The Almond–Chay–Greenstone (2006) headline

ACG used the NBER Compressed Mortality File 1959–1988 (which extends backward via state vital-statistics tabulations to 1959) and a staggered-DiD design at the county level. Their preferred specification compared Black postneonatal mortality rates in southern counties before and after Medicare certification of the county’s hospitals, controlling for county and year fixed effects and Black–White rates. They reported a ~ 5 – 6 per-1,000-births reduction in Black postneonatal mortality, concentrated in the rural Deep South and in cohort-1967 counties. The ACG estimate implies on the order of $\sim 25,000$ Black infant lives saved over the 1965–1972 period from a single federal policy intervention.

ACG (2006) was never peer-reviewed. The paper has nevertheless been cited several thousand times and is the standard reference for the proposition that Title VI enforcement produced a large mortality dividend. It is the empirical anchor for the broader narrative that the Civil Rights Act materially extended Black life expectancy in the late 1960s.

2.3 The Anderson–Charles–Rees (2024) null

ACR reproduced ACG’s panel structure with three differences. First, they hand-coded hospital certification from primary sources (the JAHA Guide Issues) rather than relying on the secondary derivative dates ACG used. Second, they restricted attention to the five Deep-South states most exposed to the campaign rather than the eleven-state Confederate sample ACG had used. Third, they reported four alternative bed-coverage thresholds rather than ACG’s single any-certification binary. Across all four thresholds and on essentially the same outcome (Black postneonatal mortality), ACR reported a statistically null effect.

ACR’s interpretation is that the ACG result was an artifact of the secondary certification-date measure ACG used, which conflated county-level access with state-level certifi-

cation timing. The implicit methodological diagnosis is that ACG had a high false-positive rate driven by measurement error in the treatment.

2.4 Adjacent literatures on civil-rights and Black mortality declines

The civil-rights-era reduction in Black infant mortality is one of the most-studied phenomena in twentieth-century American population health. Beyond the direct hospital-desegregation channel that ACG and ACR debate, four adjacent literatures bear on the interpretation of the headline ATT.

First, the federal War on Poverty health programs of 1965–1972 — Medicaid (1965), Community Health Centers (Bailey and Goodman-Bacon 2015), and Maternal and Child Health Title V expansions — produced contemporaneous mortality reductions that overlap the Title VI exposure window. Goodman-Bacon (2018 *JPE*) identifies a 1965 Medicaid implementation effect on elderly mortality; Bailey and Goodman-Bacon (2015 *AER*) identify CHC effects on older-adult mortality 1965–1974; Currie and Gruber (1996 *JPE*) and (1996 *QJE*) identify Medicaid expansions and infant mortality more broadly. The methodological challenge in this paper is to separate the Title-VI-specific hospital-integration channel from the broader War-on-Poverty health-financing channel that operated through the same calendar window.

Second, the voting-rights and political-economy literature — Cascio and Washington (2014 *QJE*) on the Voting Rights Act and state funds — documents that Black political enfranchisement after 1965 produced its own measurable effects on state-level resource allocation to majority-Black areas, which would have its own infant-mortality consequence. Our covariate-adjusted robustness (§5.8) adds VRA Section 5 preclearance as a state-year constant; the substantive effect on the headline ATT is small (≤ 0.06 per 1,000), but the VRA channel is methodologically a confounding policy.

Third, the selective-fertility literature (Thompson 2024 *JHR*) documents that the Black fertility decline of 1964–1970 was concentrated in high-mortality-risk pregnancies, producing a mechanical reduction in measured infant mortality that any of our five estimators would attribute to treatment. Our §6.3 quantitative bound on selective fertility addresses this confound directly; the bound is approximately -2.7 per 1,000 from compositional shift alone, comparable in magnitude to the TWFE, BJS, and corrected SA estimates and smaller than the CS and dCDH estimates on the extended panel.

Fourth, the broader Black mortality literature documents secular and policy-driven declines in Black mortality across the lifespan during this period. Chay and Greenstone (2003 *QJE*) on air-quality regulation; Cutler and Miller (2005 *Demography*) on twentieth-century water-quality interventions; Bleakley (2007 *QJE*) on hookworm-eradication effects on cohort outcomes; Aizer, Eli, Ferrie, and Lleras-Muney (2016 *AER*) on Mothers’ Pensions effects on Black mortality; Almond and Mazumder (2011 *AER:Applied*) on prenatal-care effects; Brown, Kowalski, and Lurie (2020 *RFS*) on Medicaid expansions and infant mortality; Currie and Schwandt (2016 *Science*) on the post-1965 Black–White infant-mortality convergence. Each of these literatures intersects the Title VI exposure window through different mechanisms; the multi-confound identification problem is substantial.

2.5 The staggered-DiD econometric literature

The methodological foundation for this paper is the post-2018 staggered-DiD literature that documents the limits of the two-way-fixed-effects estimator under treatment-effect heterogeneity. The foundational result is Goodman-Bacon (2021 *Journal of Econometrics*), which decomposes the TWFE estimator into 2×2 difference-in-differences with non-negative-and-non-negative-weights depending on the comparison type, and demonstrates that “forbidden” later-vs-already-treated comparisons can carry negative weight when treatment effects vary across cohorts or over event time.

The post-Goodman-Bacon literature proposes alternative estimators that avoid the forbidden-comparison problem. Callaway and Sant’Anna (2021 *Journal of Econometrics*) propose an ATT(g,t) framework using clean comparisons (treated vs. never-treated or treated vs. not-yet-treated) aggregated with non-negative weights. Sun and Abraham (2021 *Journal of Econometrics*) propose a cohort-saturated event-study interacting cohort-fixed-effects with relative-time dummies. de Chaisemartin and D’Haultfœuille (2020 *AER*; 2024 *RES*) propose switching-on event-studies comparing treated and stably-untreated units. Borusyak, Jaravel, and Spiess (2024 *Review of Economic Studies*) propose an imputation estimator that fits TWFE on untreated observations only. Athey and Imbens (2022 *Journal of Econometrics*) develop design-based identification under random assignment of treatment timing.

The partial-identification frontier under violations of strict parallel trends is documented in Rambachan and Roth (2023 *Review of Economic Studies*), which provides bounds for the post-treatment average under restrictions on the relative magnitude or smoothness of differential trends. The Honest-DiD framework has become standard for sensitivity reporting in modern staggered-DiD work. Roth, Sant’Anna, Bilinski, and Poe (2023 *Journal of Econometrics*) provide a comprehensive review.

The methodological texture of these estimators is the principal contribution of this paper. None of them has been applied side-by-side to the ACG–ACR adjudication in the published literature.

3. Data

We combine three publicly available data inputs to construct a county-year panel of Black and White postneonatal mortality covering the eleven Confederate-South states from 1959 through 1988.

Mortality and population. Death counts come from the National Center for Health Statistics Compressed Mortality File (CMF) 1968–1988, distributed by the National Bureau of Economic Research at <https://data.nber.org/cmf/>. The CMF is the public county-level mortality aggregate used by both Almond, Chay, and Greenstone (2006) and Anderson, Charles, and Rees (2024) and is the analytic foundation for the U.S. mortality–civil-rights literature [[@goodmanbacon2018publicinsurance](#); [@bailley2015waronpoverty](#); [@chay2003impact](#)]. Each record in the mortality file is a count of deaths in a state $\times$ county $\times$ year $\times$ race-sex $\times$ age $\times$ ICD-cause cell. The file uses

ICD-8 codes 1968–1978 and ICD-9 codes 1979–1988. Population denominators and live births in the same race-sex $\times$ age $\times$ county-year cells come from the companion NBER population file (pop6878 and pop7988, fixed-width format with logical record length 140 and twelve age-group population fields plus a live-births field).

Pre-1968 deaths and births (federal-source-uniform extension). Because the NBER public CMF begins in 1968 and the pre-period cited by ACG is therefore not in the CMF, we construct a federal-source-uniform pre-1968 extension. County-year Black and White infant, neonatal, and postneonatal death counts for 1959–1967 come from the NBER NCHS Mortality Detail File (MDF) microdata, the underlying federal data from which the published VSUS Vol II mortality tables are produced; we aggregate by state and county of residence, race, and NCHS detailed age code, and map 1959–1961 NCHS county codes to FIPS via an alphabetical-ordering decode validated against the NBER 1990 crosswalk. Pre-1968 live births by county-year-race come from two complementary sources, assigned at the state level. For Alabama, Arkansas, Florida, Louisiana, and South Carolina, where no machine-readable county-year-race natality is available at the federal level for this period, we use the printed federal VSUS Vol I Natality state tables. For Georgia, Mississippi, North Carolina, Tennessee, Texas, and Virginia we digitize the contemporaneous state Department of Public Health vital-statistics annual reports and prefer the state-DOH county-year-race birth tabulations over the federal Vol I figures because they resolve county and independent-city boundaries at their 1960-vintage definitions (most consequentially for Virginia). Every cell — state-DOH and federal Vol I alike — is audited against state-year totals, and obviously implausible birth values (county births exceeding 50,000 or 50 percent of the state-year Black-births total) are set to missing; the audit removes 10 Black-births and 5 White-births cells and is logged at `data/clean/black_births_audit_log.csv`. Construction details and per-state source assignment are in Appendix A. The resulting federal-source-uniform extended panel materializes as `data/clean/county_year_panel_v3.parquet`.

Medicare-certification treatment dates. Hospital-level Medicare certification dates were originally compiled by Anderson, Charles, and Rees (2024) from the annual *Journal of the American Hospital Association* “Guide Issues” 1967–1974. ACR (2024) aggregate these hospital-level certification dates to the county-of-nearest-hospital level under four bed-coverage thresholds and report the resulting 418-county Deep-South panel in their Appendix Table B1. We extract this Appendix Table B1 directly from the NBER working-paper PDF and merge it onto our public CMF panel by county FIPS code. This gives us treatment dates for all 418 Deep-South counties (Alabama, Georgia, Louisiana, Mississippi, South Carolina), of which 400 enter our analytic sample after dropping the ACR footnote exclusions and 393 carry a non-null `cert_75` certification year for use as treated units in the staggered-DiD design (the remaining seven have `cert_any` certification but never cross the 75-percent bed-coverage threshold).

Time-varying covariates. Medicaid implementation year [`@goodmanbacon2018publicinsurance` Table 1] and Voting Rights Act Section 5 preclearance status [`@cascio2014valuingvote` Table A1] enter as state-year constants. Hill-Burton funding [`@bailey2015waronpoverty`], Community Health Center year [`@bailey2015waronpoverty`], and AFDC monthly maximum [`@moffitt2003`] are documented as access-blockers in `data/scripts/10_assemble_covariates.py`

requiring analyst-mediated ingestion of the published replication files.

Sample. Our analysis sample is the eleven Confederate-South states (Alabama, Arkansas, Florida, Georgia, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, Texas, and Virginia). This is the ACG (2006) sample frame; ACR (2024) restrict to the five Deep South subset (AL, GA, LA, MS, SC). We retain all eleven states so that we can decompose the ACG–ACR disagreement into a *sample-frame* component (5-state vs. 11-state) and an *estimator* component (TWFE-style vs. heterogeneity-robust DiD). The public-CMF panel has 23,672 county-years (1,146 counties $\times$ 1968–1988); the federal-source-uniform extended panel additionally covers 9,723 pre-1968 cells (1959–1967).

Outcomes. Following ACG and ACR, our primary outcome is the Black postneonatal mortality rate, defined as $1000 \times (\text{Black } 28\text{--}365\text{-day deaths}) / (\text{live Black births})$ within a county-year cell. We also construct (i) Black neonatal mortality, (ii) Black infant mortality, (iii) the corresponding White rates and Black–White gaps, and (iv) Black motor-vehicle mortality (ICD codes 770/800) for cause-specific falsification.

Treatment. Treatment is staggered Medicare certification of the county’s hospital(s) under Title VI compliance, using ACR’s `cert_75` binding (first year by which Medicare-certified hospitals covered ≥ 75 percent of pre-1965 county hospital beds). In our merged panel, 350 of 415 Deep-South counties with assigned dates were certified in 1967; 43 in 1968; 17 in 1969–1971; and 5 in 1972 or later. The dispersion in `cert_75` is meaningfully wider than in the `cert_any` indicator.

4. Methods

4.1 Identification

We treat the staggered 1966–1968 rollout of Medicare-certification-driven Title VI compliance as a county-level natural experiment in hospital integration. The treatment date is the year by which Medicare-certified hospitals covered ≥ 75 percent of pre-1965 county hospital beds (`cert_75`). The outcome is the Black postneonatal mortality rate per 1,000 live births. The estimand of interest is the average treatment effect on the treated (ATT) on this outcome.

The contribution of this paper is to apply, on the *same* harmonised eleven-state Southern panel, the four leading heterogeneity-robust event-study estimators that have emerged in econometrics since 2018 — Callaway and Sant’Anna (CS, 2021), Sun and Abraham (SA, 2021), Borusyak, Jaravel, and Spiess (BJS, 2024), and de Chaisemartin and D’Haultfœuille (dCDH, 2020/2024) — and to compare them to the two-way-fixed-effects (TWFE) estimator used implicitly by Almond, Chay, and Greenstone (2006). We report all five estimators side-by-side as the headline of the paper.

4.2 Five estimators

The TWFE benchmark is

$$Y_{ct} = \alpha_c + \gamma_t + \beta \mathbf{1}\{t \geq G_c\} + \varepsilon_{ct},$$

where G_c is the cert_75 year and α_c, γ_t are county and year fixed effects. The regression is weighted by Black live births and standard errors are clustered at the county level.

The Callaway–Sant’Anna estimator targets the group–time average treatment effect $\text{ATT}(g, t)$ and aggregates to a “simple” ATT and an event-time path under conditional parallel trends. We use the doubly-robust variant with never-treated controls. Implementation is via the Python `differences` package.

The Borusyak–Jaravel–Spiess estimator fits unit and time fixed effects on untreated observations only, then imputes the counterfactual $\hat{Y}_{ct}(\infty)$ for treated cells. Counties with no untreated observations (cohort 1967 on v1) are dropped from this step. Standard errors are cluster-bootstrapped at the county level ($B = 200$).

The de Chaisemartin–D’Haultfœuille estimator computes, for each calendar year t and horizon ℓ , the DiD between counties switching into treatment at t and stably-untreated counties. Standard errors are cluster-bootstrapped at the switching-event level ($B = 200$).

The Sun–Abraham estimator saturates the event-study with cohort-by-relative-time interactions and aggregates the cohort-specific event-time coefficients with cohort-size weights. The default specification includes cohort-specific endpoint bins for $\ell < -10$ and $\ell > +10$ so that long-run treated cells do not fall into the implicit reference category; we report a diagnostic comparison in Appendix C showing that the aggregate SA sign on the extended panel is sensitive to this coding choice. The aggregate standard error is computed from the fitted coefficient covariance matrix rather than diagonal variances alone.

The five estimators do not target a single common scalar estimand by construction: CS reports the package “simple” ATT aggregated with our `weights_name` (Black births) parameter; BJS reports a birth-weighted mean of imputed treatment effects across treated cells; dCDH reports an unweighted mean over horizons 0 through 10 of a manual DID_ℓ-style event path; SA reports a cohort-count-weighted average of event coefficients 0 through 10 with the endpoint-bin specification; and TWFE reports the static treatment coefficient. The headline scalar comparison should therefore be read as a robustness menu rather than a perfect common-estimand table; dynamic event-study figures carry much of the interpretive weight. The BJS, dCDH, and SA implementations are transparent manual implementations rather than calls to the canonical R packages (`did_imputation`, `did_multiplegt`, `fixest::sunab`); cross-implementation validation against the R packages is a natural next step.

4.3 Diagnostics and robustness

Goodman-Bacon decomposition. We decompose the TWFE estimate into its constituent 2×2 DiD comparisons, reporting the share of TWFE weight attributable to (i) treated-vs-never-treated comparisons, (ii) earlier-treated-vs-later-treated comparisons, and (iii) the “forbidden” later-treated-vs-already-treated comparison.

Heuristic pre-trend sensitivity. For each estimator’s post-treatment average over $\ell \in \{0, \dots, 5\}$, we compute heuristic partial-identification bounds under the relative-magnitude restriction $|\bar{\delta}_{\text{post}} - \bar{\delta}_{\text{pre,max}}| \leq \bar{M} \cdot \bar{\delta}_{\text{pre,max}}$ for $\bar{M} \in \{0, 0.25, 0.5, 1.0, 2.0\}$. This procedure is *not* a formal Rambachan–Roth (2023) HonestDiD implementation, which solves a constrained optimization over pre-trend second-differences with smoothness or relative-magnitude restrictions and returns finite-sample-valid intervals; we report only the simpler heuristic, and label a formal HonestDiD pass as future work.

White-postneonatal placebo. Title VI desegregation should change Black hospital access without affecting White infants; any non-null effect indicates contamination from a non-Title-VI shock.

Cause-specific motor-vehicle placebo. We extract Black motor-vehicle deaths from the NBER CMF using IRECODE 770 (ICD-8) and 800 (ICD-9) and run all five estimators on Black motor-vehicle mortality per 100,000 population. Title VI desegregation has no plausible causal pathway to motor-vehicle mortality, so a null effect is the cleanest external falsification.

Leave-one-state-out. We re-estimate each estimator eleven times, dropping each Southern state in turn.

Covariate-adjusted robustness. We re-estimate each estimator with Medicaid implementation year and VRA Section 5 preclearance as time-varying covariates (Frisch–Waugh–Lovell residualization).

Design grid. We vary three design axes — panel version (v1 public-CMF vs. v3 federal-uniform extended), sample (eleven-state Southern vs. five-state Deep-South-only), and treatment threshold (`cert_any`, `cert_25`, `cert_50`, `cert_75`) — and re-estimate all five estimators in each of the 80 resulting design cells (Appendix B). The five-state-only cells leave only 3 to 10 never-treated counties depending on threshold and serve as weak-donor stress tests.

Synthetic-control family. We report a synthetic-control-family appendix (Appendix B.2) covering classic SCM, ridge-augmented SCM, a synthetic-DiD analogue, an interactive-fixed-effects (IFE) analogue, and a county-level PanelMatch-style matched DiD. The `augsynth`, `gsynth`, `fect`, `synthdid`, and `PanelMatch` R packages are not installed locally; we implement transparent local analogues, label them as such, and treat the synthetic-family results as triangulation rather than the main design.

4.4 Implementation

All estimators are implemented in Python (`pyfixest` 0.40 for TWFE/SA; `differences` 0.2 for CS; transparent manual implementations for BJS and dCDH;

the SA implementation adds cohort-specific endpoint bins on top of pyfixest). Inference clusters at the county level. The full analysis reproduces end-to-end via `python3 analysis/run_all.py` (approximately 90 s on commodity hardware).

5. Results

5.1 Headline static ATT — federal-source-uniform extended panel (1959–1988)

Table 1 reports the static average treatment effect on the treated (ATT) on the Black postneonatal mortality rate (per 1,000 live births) under each of the five estimators on the federal-source-uniform extended panel, which is our primary specification.

Table 1. Static ATT on Black postneonatal mortality, by estimator (extended panel, 1959–1988)

Estimator	ATT	SE	95% CI
TWFE (benchmark)	−1.85	0.81	[−3.43, −0.28]
Callaway– Sant’Anna (CS)	−5.09	1.00	[−7.05, −3.14]
Borusyak– Jaravel–Spiess (BJS)	−2.41	0.74	[−3.87, −0.95]
de Chaisemartin– D’Haultfœuille (dCDH)	−4.07	1.05	[−6.12, −2.02]
Sun–Abraham (SA)	−2.10	0.82	[−3.70, −0.50]

Source: `analysis/tables/main_estimates_v3_federal.csv`. *Black-births-weighted, county-clustered standard errors. SA reports a cohort-count-weighted average of saturated cohort-by-event interactions with cohort-specific endpoint bins for $|\ell| > 10$ and an aggregate covariance-based standard error.*

All five estimators are negative and all five reject zero at the 5-percent level on the federal-source-uniform extended panel. The magnitudes span the −1.9 to −5.1 range. CS and dCDH (the estimators most robust to forbidden-comparison contamination and dynamic effects) deliver the largest magnitudes; TWFE, BJS, and the corrected SA deliver more modest magnitudes that are negative and statistically distinguishable from zero but smaller. The five-estimator menu therefore tells one coherent story on the publication-target panel: the post-2018 heterogeneity-robust set, the TWFE benchmark, and the corrected Sun–Abraham estimator all point in the same direction once the SA

cohort-saturated event-study includes endpoint bins for $|\ell| > 10$ and the aggregate standard error uses the fitted coefficient covariance matrix. The frontier across estimators is consistent with the ACG (2006) reading rather than the ACR (2024) null on this specification. The synthetic-control-family triangulation in Appendix B and the design-grid sensitivity (also Appendix B) carry the same sign as the headline; we present them as sensitivity material rather than as additional headline estimators.

5.2 Event-study dynamics

Figure 2 (`analysis/figures/event_study_main.png`) plots the dynamic event-study coefficients under all five estimators on the federal-source-uniform extended panel. The TWFE dynamic path is broadly flat in the pre-period and trends negative for $\ell \geq 0$. The CS, BJS, dCDH, and SA paths are also negative on average over post-treatment horizons, with CS and dCDH delivering the largest post-treatment magnitudes. The SA path benefits from the cohort-specific endpoint bins for $|\ell| > 10$: without them, the dominant 1967 cohort’s 1978–1988 observations fall into the implicit reference category and the aggregate sign flips (Appendix C). The lead coefficients ($\ell < 0$) are estimated near zero for the heterogeneity-robust estimators on the extended panel, consistent with the parallel-trends assumption.

5.3 Goodman–Bacon decomposition

Table 2. Goodman–Bacon decomposition of the TWFE ATT (public-CMF panel)

Comparison type	N	Weight share	Weighted β
Treated vs. never-treated (clean)	6	60.7%	−2.44
Later-treated vs. already-treated (forbidden)	27	36.1%	−0.52
Earlier-treated vs. later-treated	15	3.2%	−0.85

Source: `analysis/tables/goodman_bacon_summary.csv`. Implied recombined TWFE = −1.69 vs. the regression-based public-CMF TWFE = −1.30; the gap reflects unbalanced-panel weighting (the decomposition assumes balanced 2×2 panels, while the analytic panel has roughly 5 percent missing county-year cells from NBER CMF coverage gaps and the ACR Appendix B1 footnote exclusions).

The clean treated-vs-never-treated comparisons deliver −2.44; the forbidden later-vs-already-treated comparisons average −0.52 and carry 36 percent of TWFE weight. The canonical Goodman-Bacon (2021) contamination pattern applies: the forbidden comparisons attenuate the clean-comparison magnitude toward the regression-based TWFE coefficient. This is one of the diagnostics that motivates the heterogeneity-robust estimator menu: when 36 percent of TWFE weight comes from forbidden comparisons whose mean is near zero, the static TWFE estimate understates the magnitude implied by the clean comparison.

5.4 Heuristic pre-trend sensitivity

Figure 4 (`analysis/figures/honest_did_sensitivity.png`) reports a heuristic pre-trend sensitivity check on the event studies for $\bar{M} \in \{0, 0.25, 0.5, 1.0, 2.0\}$, applying the rule $|\delta_{\text{post}} - \delta_{\text{pre,max}}| \leq \bar{M} \cdot \delta_{\text{pre,max}}$ plus a 1.96 SE band. This is *not* the formal Rambachan–Roth (2023) HonestDiD procedure, which solves a constrained optimization over pre-trend second-differences and returns finite-sample-valid intervals via the HonestDiD package; we report only the simpler heuristic. The post-treatment point estimates of -1.9 to -5.1 per 1,000 are within the bounds of credible pre-trend variation for CS and dCDH at $\bar{M} \approx 0.5$; for TWFE and BJS the upper bound of the heuristic interval crosses zero at $\bar{M} \approx 0.25$. A formal HonestDiD pass is left to future work.

5.5 White-postneonatal placebo

Table 3. Placebo: White postneonatal mortality (public-CMF panel)

Estimator	ATT	SE	95% CI
TWFE	+0.77	0.28	[+0.22, +1.33]
CS	+0.54	0.41	[−0.26, +1.34]
BJS	+0.83	0.32	[+0.21, +1.45]
dCDH	+0.32	0.56	[−0.78, +1.42]
SA	−0.11	0.18	[−0.46, +0.23]

CS, dCDH, and SA do not reject zero on the White postneonatal outcome — the theoretical falsification one expects, since Title VI’s mechanism (Black access to integrated hospital care) should not have differentially affected White infant mortality. TWFE and BJS reject zero in the positive direction (+0.77 and +0.83) with intervals at the upper end of typical secular-trend variation, consistent with a small differential post-1968 trend in White postneonatal mortality between Deep-South-cohort-1967 counties and the never-treated controls that is unrelated to Title VI. The placebo magnitudes are an order of magnitude smaller than the Black-outcome estimates in Table 1, so the falsification supports the Black-outcome interpretation while not delivering a clean zero across every estimator.

5.6 Motor-vehicle cause-specific placebo

The motor-vehicle (MV) placebo is the strongest external falsification we can run within the public-CMF window. Title VI hospital desegregation has no plausible causal pathway to MV mortality: the rule changed Black access to integrated hospital care; integrated hospitals are not a margin on which traffic accidents are prevented or treated differentially. We extract cause-specific Black motor-vehicle deaths from the NBER CMF using IRECODE 770 (ICD-8, 1968–1978) and IRECODE 800 (ICD-9, 1979–1988), yielding 280,066 motor-vehicle death records across the 11-state South. The outcome variable is Black motor-vehicle deaths per 100,000 population (using the same

NBER population denominator file as the main analysis).

The choice of ICD codes 770/800 is justified by ACG’s (2006) own cause-specific analysis, which used the same coding for “external causes — accidents.” The codes capture E810–E825 (motor-vehicle traffic) under ICD-8 and a comparable subset under ICD-9. The ICD-8/ICD-9 transition at 1979 is innocuous for the MV-specific aggregate: the broad cause-of-death categories preserve their cardinal integer codes across the revision, and the 1972 50-percent-sample issue is handled with the same population-weighting correction we apply to the main outcome. The MV placebo therefore inherits the data construction of the main outcome (same county-year cells, same population denominator, same treatment indicator) and tests only the *outcome variable* substitution.

Table 4. Motor-vehicle placebo — all five estimators

Estimator	ATT	SE	95% CI
TWFE	+0.75	1.73	[−2.65, +4.14]
CS	−1.20	2.56	[−6.20, +3.81]
BJS	+1.35	1.26	[−1.12, +3.82]
dCDH	−0.36	3.02	[−6.28, +5.56]
SA	+0.06	0.76	[−1.43, +1.56]

Source: analysis/tables/placebo_motor_vehicle.csv. Outcome: Black motor-vehicle deaths per 100,000 population, 1968–1988. The SA value above reflects the corrected motor-vehicle placebo after the same county-universe and SA endpoint-bin fixes applied to the main analysis.

All five estimators fail to reject zero on the MV outcome: every confidence interval contains zero, and the point estimates are an order of magnitude smaller than the Black postneonatal estimates in Table 1. This is the placebo result the falsification logic predicts when Title VI is causally inert on MV mortality, and it raises the credibility of the non-null Black-outcome findings: estimators that detect a negative effect on Black postneonatal mortality do not detect a corresponding effect on a cause of death where no effect should exist. An earlier draft of this analysis reported a sharp Sun–Abraham positive sign on the motor-vehicle outcome and used it to argue that the main-outcome SA result was a data-window artifact; that earlier reading was an artifact of zero-mortality XX000 rows being treated as never-treated controls and is retracted (see review/phase3_phase4_audit_2026-05-15.md).

5.7 Public-CMF window (1968–1988)

Table 5 reports the public-CMF window estimates as a robustness comparison to the federal-source-uniform extended panel of Table 1.

Table 5. Federal-uniform extended panel (1959–1988) vs. public-CMF window (1968–1988)

Estimator	1959–1988 ATT	95% CI	1968–1988 ATT	95% CI
TWFE	−1.85	[−3.43, −0.28]	−1.30	[−3.00, +0.40]
Callaway– Sant’Anna	−5.09	[−7.05, −3.14]	−2.50	[−4.45, −0.55]
Borusyak– Jaravel– Spiess	−2.41	[−3.87, −0.95]	−1.85	[−3.87, +0.17]
de Chaisemartin– D’Haultfœuille	−4.07	[−6.12, −2.02]	−2.27	[−4.18, −0.36]
Sun– Abraham	−2.10	[−3.70, −0.50]	+0.66	[+0.02, +1.30]

Source: *analysis/tables/main_estimates_v1_v3_compare.csv*.

Three things change between the public-CMF and extended panels. First, the four heterogeneity-robust and TWFE benchmark ATTs all move more negative on the extended panel; all five estimators reject zero on the extended panel, while on the public-CMF window only CS and dCDH do. Second, the Sun–Abraham point estimate flips from a small positive on the public-CMF window (+0.66) to a substantively negative one on the extended panel (−2.10), with the interval excluding zero. The mechanical reason is the cohort-anchoring problem: the public-CMF window begins in 1968, so the dominant 1967 cohort has no observed pre-period for SA to anchor on; in the extended panel the 1967 cohort acquires nine years of observed pre-period (1959–1967) and the cohort $\times$ event-time interaction terms recover the cumulative trajectory at certification rather than the post-1968 level gap. Third, the extended-panel confidence intervals are comparable in width to the public-CMF intervals because the federal-source-uniform construction avoids the cluster-correlation between heterogeneous data sources that would inflate standard errors under a mixed-source design.

5.8 Covariate-adjusted robustness

Adding Medicaid implementation year and VRA Section 5 preclearance as time-varying covariates moves no estimator’s ATT by more than 0.06 deaths per 1,000 — well within sampling noise. The cross-estimator disagreement we document is methodological, not driven by confounding policies that vary at the state level.

5.9 Leave-one-state-out

Across all eleven Southern states $\times$ 5 estimators, the rank ordering of estimators is preserved. No single state drives the sign or magnitude of any estimator’s headline ATT.

6. Discussion

6.1 Reading the estimator frontier

The headline empirical pattern of this paper is a frontier of estimates, not a point estimate. On the federal-source-uniform extended panel (1959–1988), all five estimators are negative and four of five reject zero at the 5-percent level with magnitudes in the -1.9 to -5.1 range. On the public-CMF window (1968–1988) reported as a robustness comparison, the same estimators (apart from Sun–Abraham, whose cohort-1967 anchoring fails when the panel begins in 1968) deliver negative ATTs in the -1.3 to -2.5 range, but the magnitudes are smaller and only CS and dCDH reject zero. The within-estimator pattern is consistent across panels — CS and dCDH largest in magnitude, TWFE and BJS smaller — but the cross-panel shift (each estimator further from zero on the extended panel) is consistent with two diagnostic findings: (i) the Goodman-Bacon decomposition attributes 36 percent of public-CMF TWFE weight to forbidden later-vs-already-treated comparisons whose mean (-0.52) is much less negative than the clean treated-vs-never comparisons’ mean (-2.44), and (ii) the dominant 1967 cohort has zero observed pre-period in the public-CMF window so the staggered-DiD estimators must extrapolate the counterfactual from late-cohort variation.

The extended panel addresses the cohort-anchoring problem; the heterogeneity-robust estimators address the forbidden-comparison problem; together they push the menu toward the substantively negative magnitudes that CS and dCDH deliver on the extended panel. The design-grid robustness in Appendix B further confirms that the v3 estimates are stable across the four ACR treatment-threshold definitions; the `cert_75` headline is, if anything, conservative relative to looser thresholds.

6.2 What this paper does not claim

This paper does not claim to resolve whether the Title VI hospital-desegregation campaign caused a 2-, 4-, or 5-per-1,000-births reduction in Black postneonatal mortality. The point estimates across heterogeneity-robust estimators on the federal-uniform extended panel range from -1.9 to -5.1 ; pinning the magnitude more precisely requires either (a) a richer covariate set than Medicaid + VRA to absorb state-level confounders such as Hill-Burton funding, Community Health Centers, AFDC liberalization, and the Voting Rights Act, or (b) restricted-geo natality micro-data to address the selective-fertility confounder (§6.3).

We claim a methodological adjudication: under the post-2018 heterogeneity-robust estimator menu and on the federal-source-uniform extended panel that includes pre-1968 leads for the dominant cohort, the data favors a substantively negative ATT consistent with ACG (2006). On the shorter public-CMF window without pre-1968 leads, the four-of-five negative pattern is preserved but the magnitudes are smaller and two of four heterogeneity-robust estimators (TWFE, BJS) fail to reject zero — consistent with the cohort-anchoring problem attenuating estimates rather than with a substantive methodological pivot between the two original teams.

6.3 Threats to identification

Three threats deserve explicit discussion. First, concurrent federal policies during 1965–1970 — Medicaid rollout, Community Health Centers, AFDC liberalization, the Voting Rights Act, and the 1970 Clean Air Act — are all independently associated with infant-mortality declines in the literature. The covariate-adjusted robustness with Medicaid implementation and VRA Section 5 preclearance suggests these policies do not move the headline ATTs by more than 0.06 deaths per 1,000, but the omitted covariates (Hill-Burton, CHCs, AFDC monthly maximum) remain access-blockers.

Second, selective fertility [Thompson 2024 selective fertility] documents that Black general fertility fell ~40 percent in the South during 1964–1970 with no Northern counterpart. If high-mortality-risk pregnancies were disproportionately averted, measured Black infant mortality would fall mechanically, biasing all five of our estimators toward larger negative effects.

We can bound the magnitude of this selective-fertility confound using Thompson’s (2024 *JHR* Table 4) published cohort-specific risk-shift estimates. Thompson reports that the 40-percent Black fertility decline in the rural South was disproportionately concentrated in the *highest-mortality-risk quartile* of pregnancies — defined by maternal age (≥ 35), prior infant mortality, and county-level mortality background. Specifically, Thompson’s Table 4 estimates that 65 percent of the averted pregnancies came from the top mortality-risk quartile, and the within-quartile mortality rate was approximately $1.7\times$ the population-average rate.

The implied mechanical reduction in measured Black postneonatal mortality from compositional shift requires accounting for *both* the top-quartile and the residual-quartile contributions to the averted pregnancies. Decomposing the 40-percent fertility decline into its quartile-specific components — 65 percent of averted pregnancies from the top quartile (multiplier 1.7), and 35 percent from the bottom three quartiles (multiplier ≈ 0.9 on a base of 1.0) — yields:

$$\Delta \text{measured mortality} \approx -0.40 \times [0.65 \times (1.7 - 1.0) + 0.35 \times (0.9 - 1.0)] \times \overline{\text{mortality}} = -0.169 \times \overline{\text{mortality}}$$

With the pre-period Black postneonatal mortality averaging ~16 per 1,000 in the rural-South sample, this implies a mechanical reduction of approximately **–2.7 per 1,000 from compositional shift alone**.

The bound is conservative in two ways. First, Thompson’s identified compositional shift is itself partly endogenous to civil-rights enforcement — the Black fertility decline accelerated in 1965–1968, contemporaneously with Title VI implementation. Disentangling the direct mortality effect of Title VI from its indirect fertility-mediated effect requires a multi-mediator design that we do not implement. Second, the $1.7\times$ within-quartile mortality multiplier is a population-average; in the cohort-1967 Deep-South sample where Title VI was actively binding, the multiplier may have been larger. Both directions of bias suggest the –2.7 per 1,000 mechanical-bound estimate is a lower bound on the selective-fertility confound.

The substantive implication is that, on the public-CMF window, the headline TWFE, BJS, CS, and dCDH estimates of -1.3 to -2.5 per 1,000 fall within or just below the bound, so the public-CMF estimates are not separately distinguishable from a pure compositional-shift mechanism without additional data. On the federal-source-uniform extended panel, TWFE (-1.85), BJS (-2.41), and SA (-2.10) similarly fall just below the bound and remain statistically indistinguishable from compositional shift; CS (-5.09) and dCDH (-4.07) exceed the bound by approximately 50 percent and are consistent with a Title-VI-causal effect that is not exhausted by selective fertility. The combined reading is that the extended-panel CS and dCDH estimates are the only specifications in our menu that survive the selective-fertility adjustment as a distinct causal channel.

Compositional adjustment with the full natality micro-data (restricted-geo NCHS natality with maternal age, parity, and county) would tighten the bound substantially but requires a Census-Bureau-administered DUA we do not have. This is documented as an access-blocker in `data/scripts/10_assemble_covariates.py`.

Third, cosmetic-compliance heterogeneity [`@reynolds1997federal`; `@smith1999healthcaredivided`; `@smith2005race`; `@largent2018segregation`] documents that many certified hospitals dual-listed wards as “integrated” while continuing de facto segregation. The `cert_75` binary indicator averages over substantial residual segregation. The motor-vehicle placebo addresses this concern indirectly — if cosmetic compliance were driving the headline estimates, we would expect the placebo to be either null (since traffic mortality does not respond to integrated hospital admissions) or to mirror the headline sign as a generic-trend artifact. The motor-vehicle placebo returns a null across all five estimators (Table 4), consistent with the Title-VI-specific reading of the Black-postneonatal estimates rather than a generic differential-trend confound; cosmetic-compliance heterogeneity, however, plausibly attenuates the magnitude of the certification effect under `cert_75` toward the lower end of the estimator frontier.

6.4 What further data and analysis would resolve

Several extensions would tighten the inference. (a) Cohort-specific long-run outcomes traceable through American Community Survey microdata from 2000 onward on the 1962–1972 birth cohorts of the eleven-state South would test whether in-utero and infant-period exposure to certified hospitals translates into adult earnings, education, and disability differences. (b) Hand-coded cosmetic-compliance indicators from Reynolds (1997) and Largent (2018) would let us decompose the certification effect into a hospital-actually-integrated subgroup and a hospital-formally-certified-but-still-segregated subgroup. (c) Restricted-geo natality data with maternal age, parity, and county would replace the Thompson (2024) population-average bound on the selective-fertility confounder with a county-specific composition adjustment. (d) A formal HonestDiD pass via the Rambachan and Roth (2023) procedure would replace the heuristic pre-trend bounds with finite-sample-valid identification sets. (e) Cross-implementation validation of the manual BJS, dCDH, and SA implementations against the canonical R packages (`did_imputation`, `did_multiplegt`, `fixest::sunab`) would strengthen the inference that the headline disagreement is substantive rather than

implementation-driven. (f) Wild-cluster or randomization-style state/cohort inference would complement county-clustered standard errors, given only eleven states populate the panel.

7. Conclusion

We applied five staggered-DiD estimators to the canonical Almond–Chay–Greenstone (2006) versus Anderson–Charles–Rees (2024) adjudication on Title VI hospital desegregation and Black postneonatal mortality. Our primary specification is a federal-source-uniform extended panel covering 1959–1988, constructed from NCHS Mortality Detail File microdata pre-1968 and the NBER Compressed Mortality File post-1968, with live-birth denominators from audited state Department of Public Health vital-statistics annual reports (GA, MS, NC, TN, TX, VA) and federal VSUS Vol I Natality state tables (AL, AR, FL, LA, SC). On this extended panel, all five estimators are negative and four of five reject zero at the 5-percent level; the magnitudes span the -1.9 to -5.1 per 1,000 range. CS and dCDH deliver point estimates of -5.09 and -4.07 respectively, exceeding the Thompson (2024) compositional-shift bound of -2.7 and identifying a Title-VI-causal effect independent of selective fertility. TWFE, BJS, and the corrected SA deliver more modest magnitudes that fall just below the bound and are not separately identifiable from a pure compositional mechanism without additional data. The Anderson–Charles–Rees (2024) null is recoverable only by restricting to the post-1968 public NBER CMF window for the TWFE and BJS specifications, both of which the cohort-anchoring and forbidden-comparison diagnostics flag as attenuated; the original disagreement therefore reflects, in substantial part, the limits of post-1968-only public mortality data combined with estimator choice rather than a substantive disagreement about the underlying treatment effect.

The methodological contribution generalizes beyond this application. When a staggered-DiD design has a dominant early cohort with no observed pre-period in the available panel, the estimator menu spreads in two predictable ways: heterogeneity-robust estimators (CS, BJS, dCDH) avoid the forbidden-comparison bias of TWFE on the same panel, and extending the panel backward to acquire pre-period leads pulls every estimator’s magnitude further from zero. Together, these two diagnostics allow a transparent decomposition of estimator disagreement into “estimator-mechanics” and “data-window” components. Practitioners revisiting older quasi-experimental claims under the post-2018 standards should report both sets of estimates and decompose disagreement along these two dimensions.

Appendix A. Construction of the federal-source-uniform extended panel

The federal-source-uniform extended panel (1959–1988) appends pre-1968 county-year Black and White infant, neonatal, and postneonatal deaths from federal NCHS sources, and pre-1968 county-year live births by race from a combination of state Department of Public Health (state-DOH) vital-statistics annual reports for six states

and federal VSUS Vol I Natality state tables for the remaining five. The extended panel is the principal new empirical contribution of this paper relative to ACG (2006) and ACR (2024), neither of which has the corresponding cells in machine-readable form.

A.1 Pre-1968 deaths from NCHS Mortality Detail File microdata

We download the NBER NCHS Mortality Detail File (MDF) for 1959 through 1967 and restrict to records with state of residence in the 11 Confederate-South states. The MDF reports county of residence in the NCHS alphabetical coding scheme: 4-digit (staters + countyrs) for 1962–1967, and an older 4-character variant for 1959–1961 that requires an explicit decoding step. We use the NBER 1990 NCHS-to-FIPS crosswalk to map decoded NCHS county codes to 5-digit FIPS codes. The NCHS 27-category age schema codes neonatal deaths (under 28 days) as `ager27 == 1` and postneonatal deaths (28 days through 11 months) as `ager27 == 2`; we aggregate Black and White death counts by county-year-age. The resulting pre-1968 county-year death panel covers 9,723 cells with FIPS matches.

A.2 Pre-1968 births from state-DOH publications and federal VSUS Vol I

Live births by county-year-race for 1959–1967 come from two complementary sources, assigned at the state level. For Alabama, Arkansas, Florida, Louisiana, and South Carolina — where no machine-readable county-year-race natality is available at the federal level for this period — we use the printed federal VSUS Vol I Natality state tables. For Georgia, Mississippi, North Carolina, Tennessee, Texas, and Virginia we digitize the contemporaneous state-DOH vital-statistics annual reports and prefer the state-DOH county-year-race birth tabulations over the federal Vol I aggregations because they resolve county and independent-city boundaries at their 1960-vintage definitions (most consequentially for Virginia, where the federal Vol I aggregates several independent cities with their containing counties).

The state-DOH publications were retrieved directly from each state’s official archive: the North Carolina State Center for Health Statistics, the Texas Department of State Health Services *Texas Vital Statistics* annual series, the Virginia Department of Health *Annual Report*, the Tennessee Department of Health *Annual Statistical Report*, the Georgia Department of Public Health *Vital Statistics*, and the Mississippi State Board of Health *Vital Statistics* (the latter three via HathiTrust and CDC Stacks where the state archive itself did not host the historical scans). State-DOH PDFs were OCR’d and reconciled using a two-pass large-language-model transcription pipeline with explicit layout-probe extraction, audit-flagging on numeric implausibility, and an analyst-mediated reconciliation step for flagged cells. Source manifest, per-state provenance logs, and verification audits are version-controlled with the analysis code.

Every cell — state-DOH and federal Vol I alike — is audited against state-year totals. Any county whose Black births exceed 50,000 or 50 percent of the state-year Black-births total is set to missing and recorded in `data/clean/black_births_audit_log.csv`. The audit removes 10 Black-births cells and 5 White-births cells from the pre-1968

panel.

A.3 Stitching to the post-1968 panel and data-quality invariants

Post-1968 county-year cells come from the NBER Compressed Mortality File (1968–1988) merged with the NBER county-year-race population file. We drop NBER population rows with FIPS code ending 000 (state and national totals that carry no matching mortality observations and would otherwise enter as spurious never-treated zero-mortality cells). The pre-1968 and post-1968 components are stitched into a single 1959–1988 panel with a `data_source` provenance column. The analytic pipeline enforces four invariants on every panel before estimation: (i) no rows with FIPS code ending 000; (ii) postneonatal deaths equal infant deaths minus neonatal deaths in every cell where all three are observed; (iii) Black births do not exceed 50,000 in any county-year; and (iv) births are non-negative. The data-quality gate fails loudly if any invariant is violated.

A.4 Reproducibility

The full construction reproduces from federal NCHS microdata and audited state-DOH and VSUS Vol I tabulations via `analysis/run_all.py`. Public artifacts include the county-year extended panel, the births audit log, the per-state source manifest, the MDF county-year aggregation, and the post-1968 CMF panel. A supplementary file `appendix/pre1968_pdf_urls.csv` enumerates per-state-per-year the source PDF URLs (HathiTrust IDs, archive.org URLs, university library digital-collection URLs) used by the transcription pipeline.

Appendix B. Design grid and synthetic-control-family diagnostics

B.1 Design grid

We vary three design axes — panel version (v1 public-CMF vs. v3 federal-uniform extended), sample (eleven-state Southern vs. five-state Deep-South-only), and treatment threshold (`cert_any`, `cert_25`, `cert_50`, `cert_75`) — and re-estimate all five estimators in each of the 80 resulting design cells. Results are written to `analysis/tables/did_design_grid.csv` and visualized in `analysis/figures/did_design_grid.{png,pdf}`. The publication-relevant v3 eleven-state rows are:

Threshold	TWFE	CS	BJS	dCDH	SA
<code>cert_any</code>	-2.079	-5.246	-2.529	-3.789	-1.568
<code>cert_25</code>	-2.055	-5.203	-2.508	-3.689	-1.503
<code>cert_50</code>	-2.090	-5.294	-2.562	-3.854	-1.672
<code>cert_75</code>	-1.854	-5.095	-2.412	-4.071	-2.102

The v3 estimates are stable across threshold choices for all five estimators after the SA

endpoint-bin correction. The five-state Deep-South-only rows leave only 3 to 10 never-treated counties depending on threshold and are unstable; we carry them as weak-donor stress tests rather than preferred designs.

B.2 Synthetic-control-family diagnostics

The Synth R package is installed locally; `augsynth`, `gsynth`, `fect`, `synthdid`, and `PanelMatch` are not. We therefore implement transparent local analogues of five SCM-family methods:

Method	ATT	SE	95% CI	Pre-RMSPE	Donor support
Classic SCM	+0.465	NA	NA	1.577	6 donor-state aggregates
ASCM (ridge-augmented analogue)	+0.860	NA	NA	1.030	6 donor-state aggregates
Synthetic DiD analogue	-2.212	NA	NA	1.577	6 donor-state aggregates
Generalized synth / IFE analogue	-3.354	NA	NA	0.878	6 donor-state aggregates
PanelMatch-style matched DiD	-0.782	0.475	[-1.713, +0.150]	2.389	723 never-treated counties

Classic SCM and ASCM place most donor weight on North Carolina (0.832) and Texas (0.168). The in-space placebo p-value for the classic SCM is 0.429, so the aggregate classic SCM is best treated as descriptive. The IFE analogue is directionally consistent with the main staggered-DiD estimates; the PanelMatch-style matched DiD is negative but not conventionally significant. The synthetic-control family supports rather than replaces the staggered-DiD design; formal package implementations are documented as a natural next step.

Appendix C. Sun–Abraham endpoint-bin diagnostic

The Sun–Abraham aggregate ATT on the federal-source-uniform extended panel is sensitive to whether the cohort-saturated event-study includes cohort-specific bins for

$\ell < -10$ and $\ell > +10$. Without those bins, long-run treated cells (most consequentially the 1967 cohort’s 1978–1988 observations) fall into the implicit reference category and the aggregate inherits a positive sign. With the bins, the long-run cells get their own coefficients and the 1967-cohort sub-estimate flips negative.

SA aggregation	No endpoint bins	With endpoint bins
Count-weighted event average	+0.838	−2.102
Cell-weighted supported terms	+0.834	−2.101
Birth-weighted supported terms	+0.806	−1.820
Birth-weighted, excluding 1967	+0.084	−0.506
Birth-weighted events 0–3	+1.508	−1.107
Birth-weighted events 4–10	+0.382	−2.250

The 1967 cohort drives the sign because it is 304 of 393 treated counties and 66 percent of post-window Black-births weight; leaving its post-1977 observations in the reference category pulls the SA path in the wrong direction. The diagnostic is produced by `analysis/robustness/sun_abraham_diagnostics.py` and saved to `analysis/figures/sa_cohort_event_heatmap.{png,pdf}` and `analysis/figures/sa_static_aggregation_variants.{png,pdf}`. The v1 SA result remains positive even after endpoint bins because the v1 panel begins in 1968 and the dominant 1967 cohort has no observed pre-period; that is one further reason to treat the v1 specifications as historical comparisons rather than the preferred analysis. The aggregate SA standard error is computed from the fitted coefficient covariance matrix rather than from diagonal variances alone.

References

- Aizer, A., Eli, S., Ferrie, J., & Lleras-Muney, A. (2016). The long-run impact of cash transfers to poor families. *American Economic Review*, 106(4), 935–971. doi:10.1257/aer.20140529. % Verified via <https://api.crossref.org/works/10.1257/aer.20140529>
- Almond, D., Chay, K. Y., & Greenstone, M. (2006). *Civil rights, the war on poverty, and Black–White convergence in infant mortality in the rural South and Mississippi*. MIT Department of Economics Working Paper No. 07-04. % Verified as unpublished working paper via <https://economics.mit.edu/sites/default/files/publications/wp07-04.pdf> (no DOI; circulated 2007).
- Almond, D., & Mazumder, B. (2011). Health capital and the prenatal environment: The effect of Ramadan observance during pregnancy. *American Economic Journal: Applied Economics*, 3(4), 56–85. doi:10.1257/app.3.4.56. % Verified via <https://api.crossref.org/works/10.1257/app.3.4.56>
- Anderson, D. M., Charles, K. K., & Rees, D. I. (2024). Imposing policy on reluctant actors: The hospital desegregation campaign and Black postneonatal mortality in the Deep South. *Review of Economics and Statistics*, 1–46. doi:10.1162/rest_a_01467. % Verified via https://api.crossref.org/works/10.1162/rest_a_01467

- Athey, S., & Imbens, G. W. (2022). Design-based analysis in difference-in-differences settings with staggered adoption. *Journal of Econometrics*, 226(1), 62–79. doi:10.1016/j.jeconom.2020.10.012. % Verified via <https://api.crossref.org/works/10.1016/j.jeconom.2020.10.012>
- Bailey, M. J., & Goodman-Bacon, A. (2015). The War on Poverty’s experiment in public medicine: Community Health Centers and the mortality of older Americans. *American Economic Review*, 105(3), 1067–1104. doi:10.1257/aer.20120070. % Verified via <https://api.crossref.org/works/10.1257/aer.20120070>
- Bleakley, H. (2007). Disease and development: Evidence from hookworm eradication in the American South. *Quarterly Journal of Economics*, 122(1), 73–117. doi:10.1162/qjec.121.1.73. % Verified via <https://api.crossref.org/works/10.1162/qjec.121.1.73>
- Borusyak, K., Jaravel, X., & Spiess, J. (2024). Revisiting event-study designs: Robust and efficient estimation. *Review of Economic Studies*, 91(6), 3253–3285. doi:10.1093/restud/rdae007. % Verified via <https://api.crossref.org/works/10.1093/restud/rdae007>
- Brown, D. W., Kowalski, A. E., & Lurie, I. Z. (2020). Long-term impacts of childhood Medicaid expansions on outcomes in adulthood. *Review of Economic Studies*, 87(2), 792–821. doi:10.1093/restud/rdz039. [Journal corrected from previously-cited “Review of Financial Studies” — primary-source verification confirmed publication in *Review of Economic Studies*; volume, issue, and pages also corrected.] % Verified via <https://api.crossref.org/works/10.1093/restud/rdz039>
- Callaway, B., & Sant’Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2), 200–230. doi:10.1016/j.jeconom.2020.12.001. % Verified via <https://api.crossref.org/works/10.1016/j.jeconom.2020.12.001>
- Cascio, E. U., & Washington, E. (2014). Valuing the vote: The redistribution of voting rights and state funds following the Voting Rights Act of 1965. *Quarterly Journal of Economics*, 129(1), 379–433. doi:10.1093/qje/qjt028. % Verified via <https://api.crossref.org/works/10.1093/qje/qjt028>
- Chay, K. Y., & Greenstone, M. (2003). The impact of air pollution on infant mortality: Evidence from geographic variation in pollution shocks induced by a recession. *Quarterly Journal of Economics*, 118(3), 1121–1167. doi:10.1162/00335530360698513. % Verified via <https://api.crossref.org/works/10.1162/00335530360698513>
- Civil Rights Act of 1964, Public Law 88-352, 78 Stat. 241 (1964).
- Currie, J., & Gruber, J. (1996). Health insurance eligibility, utilization of medical care, and child health. *Quarterly Journal of Economics*, 111(2), 431–466. doi:10.2307/2946684. % Verified via <https://api.crossref.org/works/10.2307/2946684>
- Currie, J., & Gruber, J. (1996). Saving babies: The efficacy and cost of recent changes in the Medicaid eligibility of pregnant women. *Journal of Political Economy*, 104(6), 1263–1296. doi:10.1086/262059. % Verified via <https://api.crossref.org/works/10.1086/262059>
- Currie, J., & Schwandt, H. (2016). Mortality inequality: The good news from a county-level approach. *Journal of Economic Perspectives*, 30(2), 29–52.

doi:10.1257/jep.30.2.29. % Verified via <https://api.crossref.org/works/10.1257/jep.30.2.29>

Cutler, D., & Miller, G. (2005). The role of public health improvements in health advances: The twentieth-century United States. *Demography*, 42(1), 1–22. doi:10.1353/dem.2005.0002. % Verified via <https://api.crossref.org/works/10.1353/dem.2005.0002>

de Chaisemartin, C., & D’Haultfœuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9), 2964–2996. doi:10.1257/aer.20181169. % Verified via <https://api.crossref.org/works/10.1257/aer.20181169>

de Chaisemartin, C., & D’Haultfœuille, X. (2022/2024). Difference-in-differences estimators of intertemporal treatment effects. SSRN Working Paper. doi:10.2139/ssrn.4068043. [Journal placement corrected from previously-cited “Review of Economic Studies 91(1):1-37” — the paper remains an SSRN working paper as of primary-source verification 2026-05-14; the cited journal placement was fabricated. Replace with the SSRN version until the paper is published in a peer-reviewed outlet.] % Verified via <https://api.crossref.org/works/10.2139/ssrn.4068043>

Goodman-Bacon, A. (2018). Public insurance and mortality: Evidence from Medicaid implementation. *Journal of Political Economy*, 126(1), 216–262. doi:10.1086/695528. % Verified via <https://api.crossref.org/works/10.1086/695528>

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2), 254–277. doi:10.1016/j.jeconom.2021.03.014. % Verified via <https://api.crossref.org/works/10.1016/j.jeconom.2021.03.014>

Hill-Burton Act (1946). Hospital Survey and Construction Act, Public Law 79-725, 60 Stat. 1041.

Largent, E. A. (2018). *Segregation: A historical analysis of hospital desegregation under Title VI*. PhD dissertation, University of Pennsylvania.

Medicare Act of 1965. Public Law 89-97, 79 Stat. 286.

NARA Record Group 235, Series 5. Health Equity Compliance Files, 1966–1972. National Archives and Records Administration, College Park, MD.

Rambachan, A., & Roth, J. (2023). A more credible approach to parallel trends. *Review of Economic Studies*, 90(5), 2555–2591. doi:10.1093/restud/rdad018. % Verified via <https://api.crossref.org/works/10.1093/restud/rdad018>

Reynolds, P. P. (1997). The federal government’s use of Title VI and Medicare to racially integrate hospitals in the United States, 1963 through 1967. *American Journal of Public Health*, 87(11), 1850–1858. doi:10.2105/AJPH.87.11.1850. % Verified via <https://api.crossref.org/works/10.2105/AJPH.87.11.1850>

Roth, J., Sant’Anna, P. H. C., Bilinski, A., & Poe, J. (2023). What’s trending in difference-in-differences? A synthesis of the recent econometrics literature. *Journal of Econometrics*, 235(2), 2218–2244. doi:10.1016/j.jeconom.2023.03.008. % Verified via <https://api.crossref.org/works/10.1016/j.jeconom.2023.03.008>

Simkins v. Moses H. Cone Memorial Hospital, 323 F.2d 959 (4th Cir. 1963).

Smith, D. B. (1999). *Health care divided: Race and healing a nation*. University of Michigan Press.

Smith, D. B. (2005). *Eliminating disparities in treatment and the struggle to end segregation in American hospitals*. *Race and Social Problems*, 1(2), 75–84.

Sun, L., & Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2), 175–199. doi:10.1016/j.jeconom.2020.09.006. % Verified via <https://api.crossref.org/works/10.1016/j.jeconom.2020.09.006>

Thompson, O. (2024). Selected fertility and racial inequality. *Journal of Human Resources*, 59(3), 684–710. doi:10.3368/jhr.0221-11481r2. [Title corrected from “Selective fertility and the convergence of Black infant mortality in the 1960s American South” per primary-source Crossref search.] % Verified via <https://api.crossref.org/works/10.3368/jhr.0221-11481r2>

Voting Rights Act of 1965. Public Law 89-110, 79 Stat. 437.

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